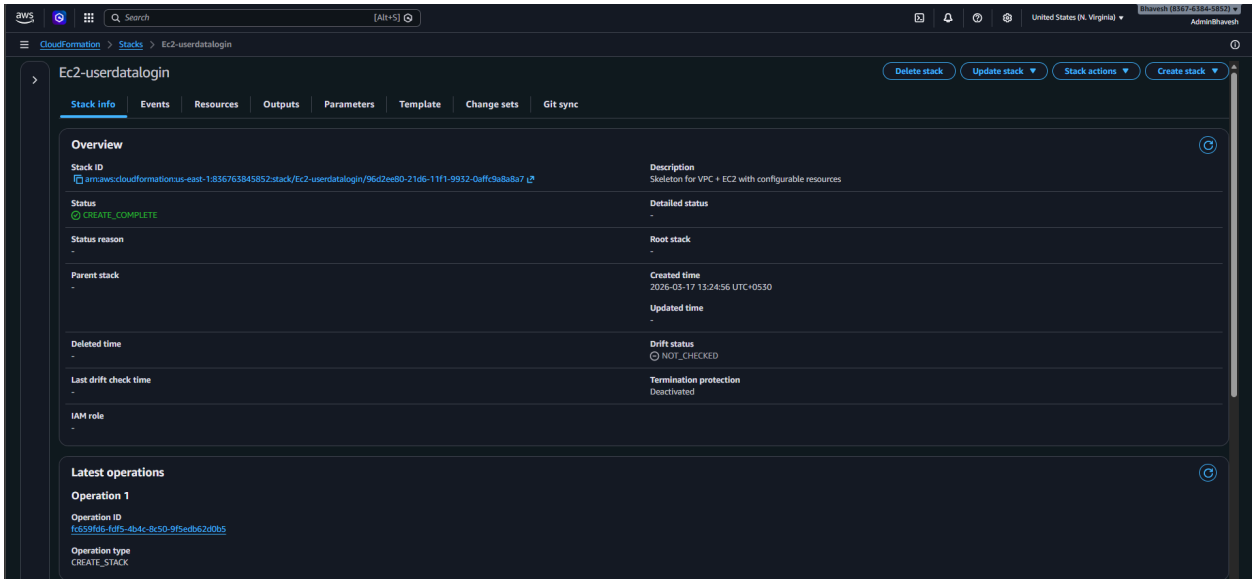
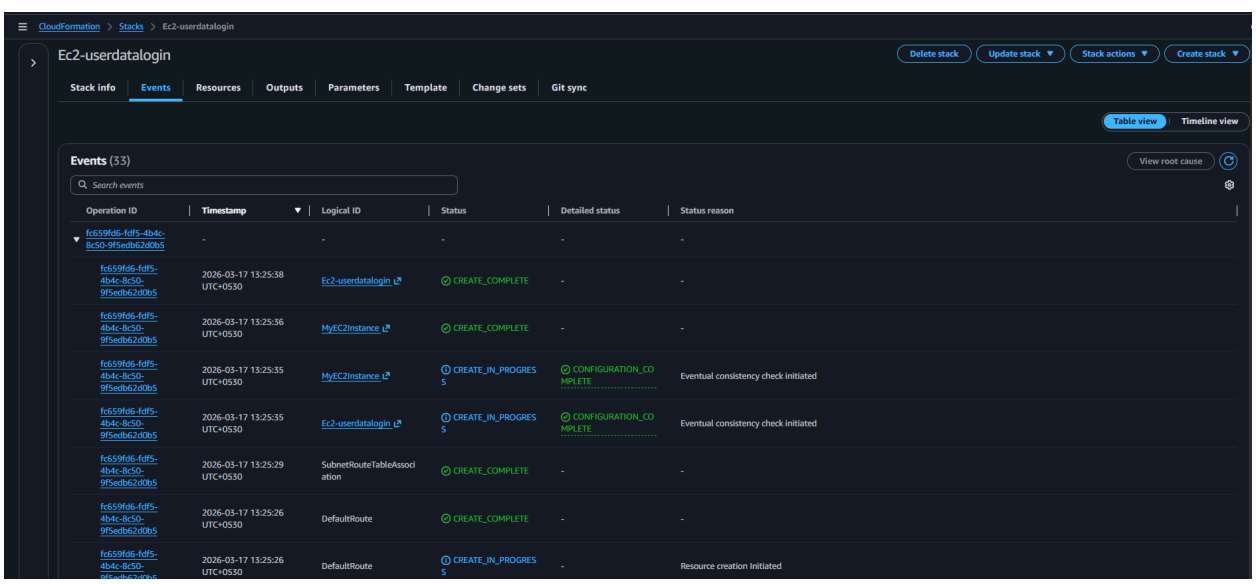
CFT

EC2 - User data Login

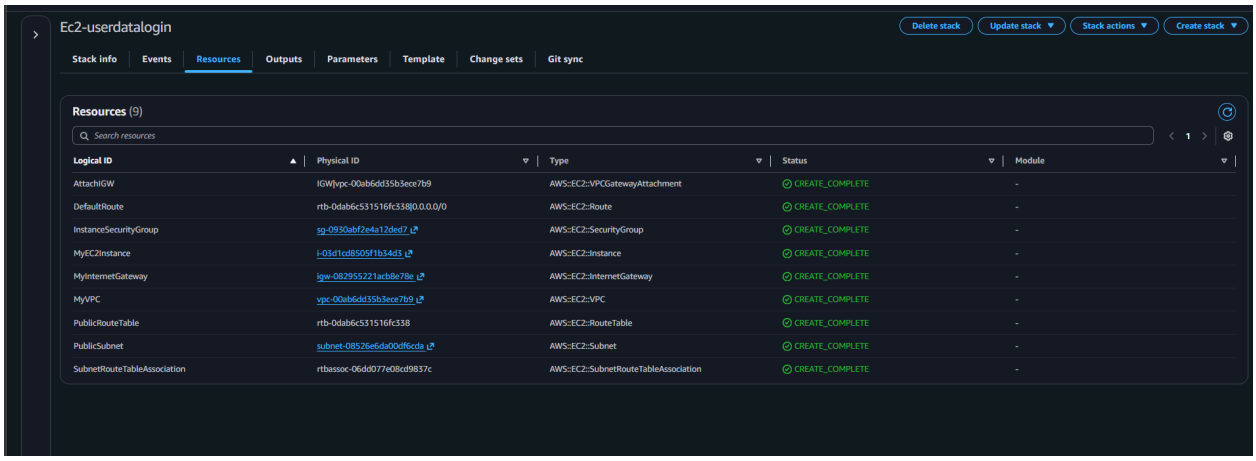
Stack



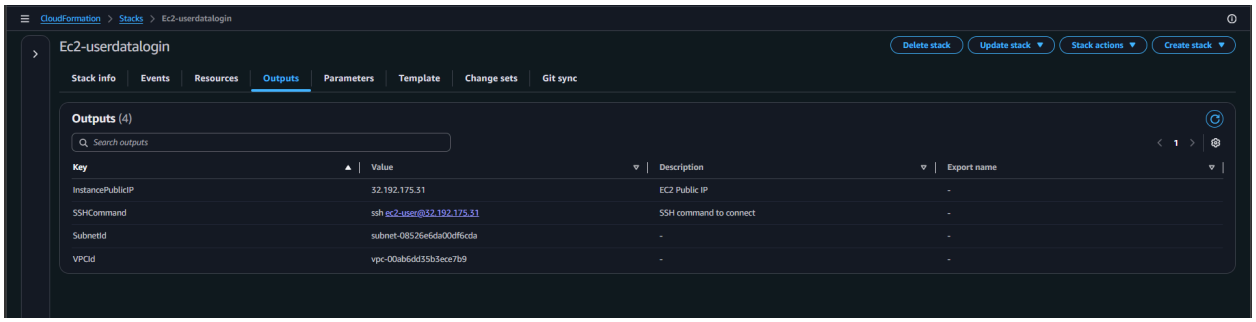
Events



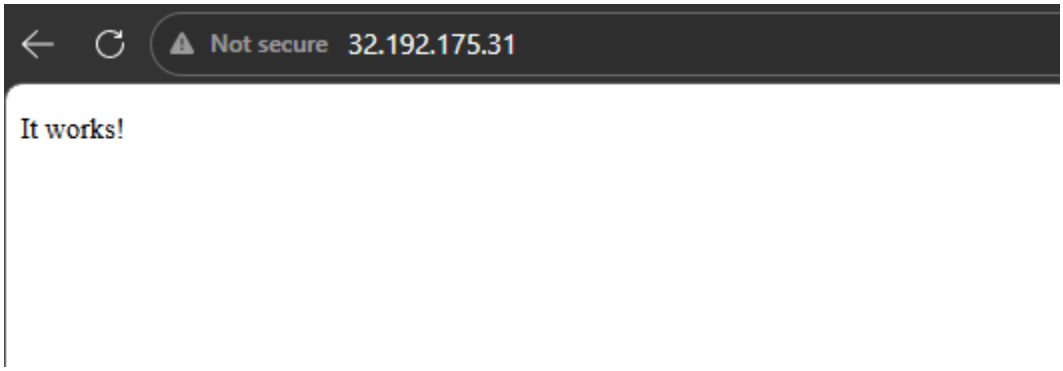
Resources



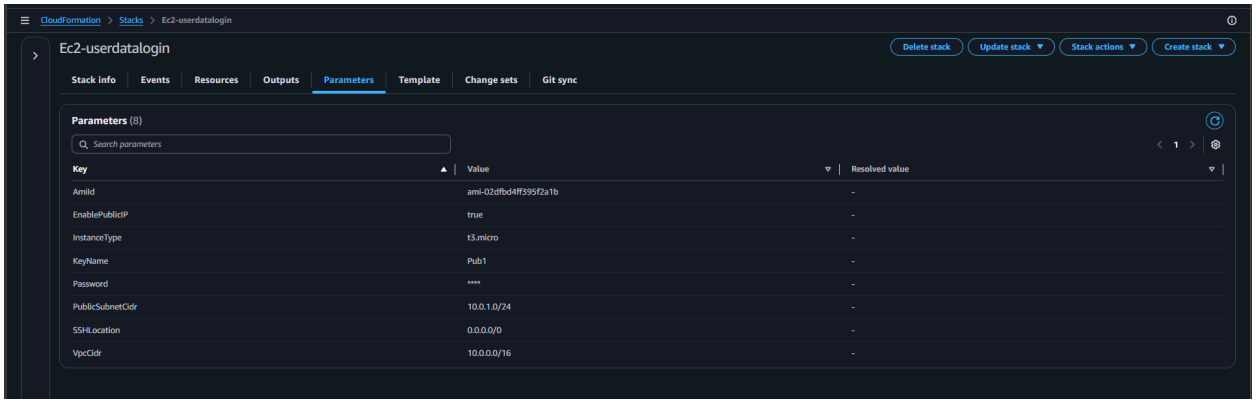
Output



HTTP



Parameters



Template

```
AWSTemplateFormatVersion: "2010-09-09"
Description: Skeleton for VPC + EC2 with configurable resources

# ========================
# PARAMETERS (CUSTOMIZE)
# ========================
Parameters:

  VpcCidr:
    Type: String
    Default: 10.0.0.0/16

  PublicSubnetCidr:
    Type: String
    Default: 10.0.1.0/24

  InstanceType:
    Type: String
    Default: t3.micro

  KeyName:
    Type: AWS::EC2::KeyPair::KeyName
    Description: Existing EC2 KeyPair
```

```yaml
  AmiId:
    Type: AWS::EC2::Image::Id
    Description: AMI ID
    Default: ami-02dfbd4ff395f2a1b

  EnablePublicIP:
    Type: String
    Default: "true"
    AllowedValues: ["true", "false"]

  SSHLocation:
    Type: String
    Default: 0.0.0.0/0

  Password:
    Type: String
    NoEcho: true
    Description: Password to set for ec2-user

# ========================
# VPC RESOURCES
# ========================
Resources:

  # ---- VPC ----
  MyVPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: !Ref VpcCidr
      EnableDnsSupport: true
      EnableDnsHostnames: true
      Tags:
        - Key: Name
          Value: MyVPC

  # ---- Internet Gateway ----
  MyInternetGateway:
    Type: AWS::EC2::InternetGateway

  AttachIGW:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref MyVPC
      InternetGatewayId: !Ref MyInternetGateway
```

```yaml
# ---- Subnet ----
PublicSubnet:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref MyVPC
    CidrBlock: !Ref PublicSubnetCidr
    MapPublicIpOnLaunch: !Ref EnablePublicIP
    AvailabilityZone: !Select [0, !GetAZs ""]
    Tags:
      - Key: Name
        Value: PublicSubnet

# ---- Route Table ----
PublicRouteTable:
  Type: AWS::EC2::RouteTable
  Properties:
    VpcId: !Ref MyVPC

DefaultRoute:
  Type: AWS::EC2::Route
  DependsOn: AttachIGW
  Properties:
    RouteTableId: !Ref PublicRouteTable
    DestinationCidrBlock: 0.0.0.0/0
    GatewayId: !Ref MyInternetGateway

SubnetRouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PublicSubnet
    RouteTableId: !Ref PublicRouteTable

# ---- Security Group ----
InstanceSecurityGroup:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: Allow SSH + HTTP
    VpcId: !Ref MyVPC
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 22
        ToPort: 22
        CidrIp: !Ref SSHLocation
```

```
        - IpProtocol: tcp
          FromPort: 80
          ToPort: 80
          CidrIp: 0.0.0.0/0

# ========================
# EC2 INSTANCE
# ========================
  MyEC2Instance:
    Type: AWS::EC2::Instance
    Properties:
      InstanceType: !Ref InstanceType
      KeyName: !Ref KeyName
      ImageId: !Ref AmiId
      SubnetId: !Ref PublicSubnet
      SecurityGroupIds:
        - !Ref InstanceSecurityGroup

      Tags:
        - Key: Name
          Value: MyEC2

      # ========================
      # USER DATA (PASSWORD ENABLE)
      # ========================
      UserData:
        Fn::Base64: !Sub |
          #!/bin/bash
          yum update -y

          # Set password
          echo "ec2-user:${Password}" | chpasswd

          # Enable password authentication
          sed -i 's/PasswordAuthentication no/PasswordAuthentication yes/g' /etc/ssh/sshd_config
          sed -i 's/#PasswordAuthentication yes/PasswordAuthentication yes/g'
/etc/ssh/sshd_config

          # Allow root login (optional)
          sed -i 's/PermitRootLogin no/PermitRootLogin yes/g' /etc/ssh/sshd_config

          # Restart SSH
          systemctl restart sshd
```

```
        # Install basic tools
        yum install -y httpd
        systemctl start httpd
        systemctl enable httpd

# ========================
# OUTPUTS
# ========================
Outputs:

  InstancePublicIP:
    Description: EC2 Public IP
    Value: !GetAtt MyEC2Instance.PublicIp

  VPCId:
    Value: !Ref MyVPC

  SubnetId:
    Value: !Ref PublicSubnet

  SSHCommand:
    Description: SSH command to connect
    Value: !Sub "ssh ec2-user@${MyEC2Instance.PublicIp}"
```